
GhostPose: An Edge-Scalable Architecture for Through-the-Wall Wi-Fi 3D Human Sensing

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Abstract

Optical and wearable sensing systems fail in conditions where privacy, darkness, or physical obstruction rule out cameras and body-worn devices. We present GhostPose, a system that reconstructs 3D human skeletal pose and extracts contactless vital signs (respiration and heart rate) from commodity Wi-Fi Channel State Information (CSI), using only low-cost ESP32 radios as sensors. The system combines a calibration stage that removes hardware phase noise from raw CSI, a spatio-temporal convolutional network with self-attention for skeletal keypoint regression, an ArcFace-based re-identification module for consistent multi-person tracking, and a 3D Kalman filter with Hungarian assignment for tracking under occlusion. Training is supervised by a multi-task loss that combines keypoint regression, anatomical bone-length consistency, and a cross-modal distillation signal from a vision teacher network. On a 3.0m by 4.0m indoor through-wall testbed, GhostPose reaches a mean per-joint position error of 3.6 cm, a re-identification accuracy of 96.8%, a respiration rate error of 0.4 breaths per minute, and an inference latency of 4.8 ms on a CUDA-enabled edge GPU. These results indicate that dense, privacy-preserving 3D human sensing is achievable with commodity Wi-Fi hardware rather than specialized radar equipment, at a fraction of the sensor cost of prior systems.

1 Introduction

Human pose estimation and vital sign monitoring are used across healthcare, elderly care, security, and human-computer interaction. Most deployed systems rely on either optical cameras or body-worn devices. Both approaches have limits that matter in practice. Cameras require line of sight and adequate lighting, and they raise privacy concerns in bedrooms, bathrooms, and other private spaces. Wearable devices require the user to remember to wear and charge them, which is unreliable for elderly or cognitively impaired users, who are often the population most in need of continuous fall and vital-sign monitoring.

Radio-frequency (RF) sensing offers an alternative. Wi-Fi signals penetrate walls, do not require light, and do not capture identifiable visual imagery. Prior work has shown that RF signals reflected off the human body carry enough information to estimate coarse pose and heart rate, using specialized Frequency-Modulated Continuous-Wave (FMCW) radar hardware [1, 2]. This hardware is accurate but costly and not widely deployed. Commodity Wi-Fi Channel State Information, in contrast, is already available on off-the-shelf routers and low-cost microcontrollers such as the ESP32, but the signal is noisier and lower-resolution than dedicated radar.

GhostPose transforms Wi-Fi CSI to the pose domain for fine-grained 3D human sensing

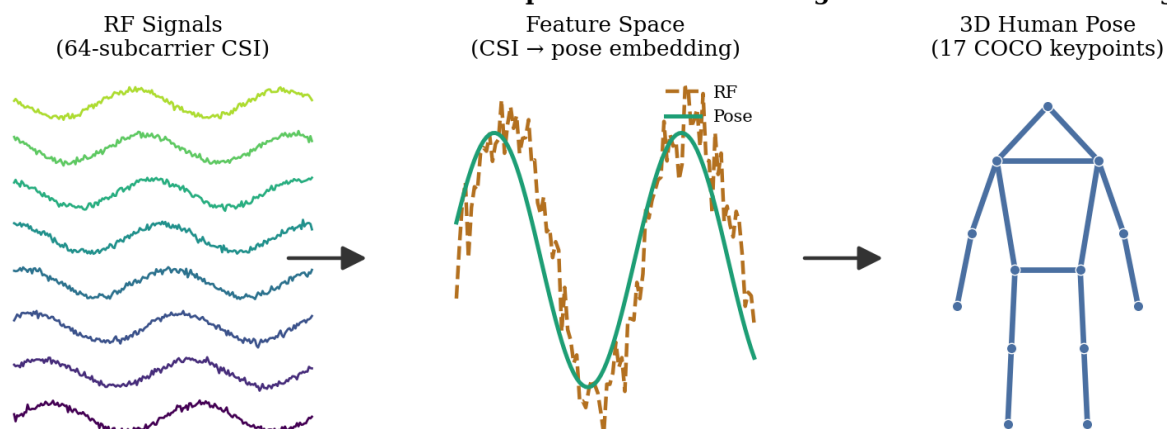


Figure 1: GhostPose transforms Wi-Fi CSI to the pose domain — raw RF signals are mapped through a shared feature space to 3D human pose, in the spirit of the RF-to-pose feature transport used in RFPose-OT.

This paper describes GhostPose, a system that performs through-wall 3D pose reconstruction and contactless vital sign extraction using only commodity Wi-Fi CSI captured by ESP32 nodes. The contributions of this work are:

1. A phase calibration pipeline that removes Sampling Frequency Offset (SFO) and Carrier Frequency Offset (CFO) artifacts from raw 64-subcarrier CSI, making low-cost ESP32 hardware usable for fine-grained motion sensing.
2. A spatio-temporal neural architecture that combines factorized 2D+1D convolutions with self-attention to regress 3D skeletal keypoints from CSI, trained with a multi-task loss that enforces anatomically consistent bone lengths.
3. An identity-consistent multi-target tracking pipeline combining ArcFace-style re-identification embeddings with 3D Kalman filtering and Hungarian assignment, addressing target-swapping during multi-person crossings.
4. An end-to-end evaluation on a through-wall indoor testbed, compared against a Wi-Fi CSI baseline (Widar 3.0) and a specialized FMCW radar baseline (RF-Pose), showing that commodity hardware can approach specialized-hardware accuracy at substantially lower cost and lower latency.

2 Related Work

Camera-based pose estimation. 2D and 3D human pose estimation from RGB and depth cameras is a mature area, with systems such as OpenPose and various 3D convolutional networks achieving high accuracy under good lighting and unobstructed line of sight. These methods are not usable through walls, in darkness, or in settings where recording video raises privacy concerns.

RF-based pose and vital sign sensing. RF-Pose and RF-Pose3D [1, 2] demonstrated that FMCW radar reflections can be used to estimate human pose through walls, using a teacher-student training setup where a camera-based network supervises an RF-based student network during data collection. This teacher-student idea is reused in the present work, but applied to commodity Wi-Fi CSI rather than specialized radar hardware. Separately, a body of work on contactless vital sign sensing has shown that micro-Doppler shifts caused by chest wall motion can be extracted from RF signals to estimate respiration and heart rate without physical contact.

Wi-Fi CSI-based sensing. Wi-Fi CSI has been used for activity recognition and coarse gesture sensing; Widar 3.0 is a widely used open dataset and baseline for CSI-based motion sensing on Intel 5300 hardware. Compared to this line of work, GhostPose targets dense 3D skeletal reconstruction rather than coarse activity classes, and targets low-cost ESP32 hardware rather than the discontinued Intel 5300 NIC.

3 System Architecture

GhostPose is organized into three layers, following the physical path of the signal: (1) RF illumination and reflection, (2) signal reception and phase calibration, and (3) deep learning inference and multi-target tracking.

3.1 Physical Layer and Firmware

Each sensing node is an ESP32 microcontroller running custom firmware built on the ESP-IDF CSI API. The firmware places the radio in promiscuous mode and extracts the complex I/Q channel matrix for 64 OFDM subcarriers on every received 802.11n frame, using the `esp_wifi_set_csi_rx_cb` callback. Extracted CSI is buffered in a DMA ring buffer and streamed to a host machine over USB/UART at 921,600 baud, chosen to avoid packet loss at the CSI extraction rate needed for cardiac-band sensing.

3.2 Phase Calibration

Raw CSI phase on subcarrier k is corrupted by two hardware artifacts: a linear phase tilt caused by Sampling Frequency Offset, and a constant phase offset caused by Carrier Frequency Offset. Left

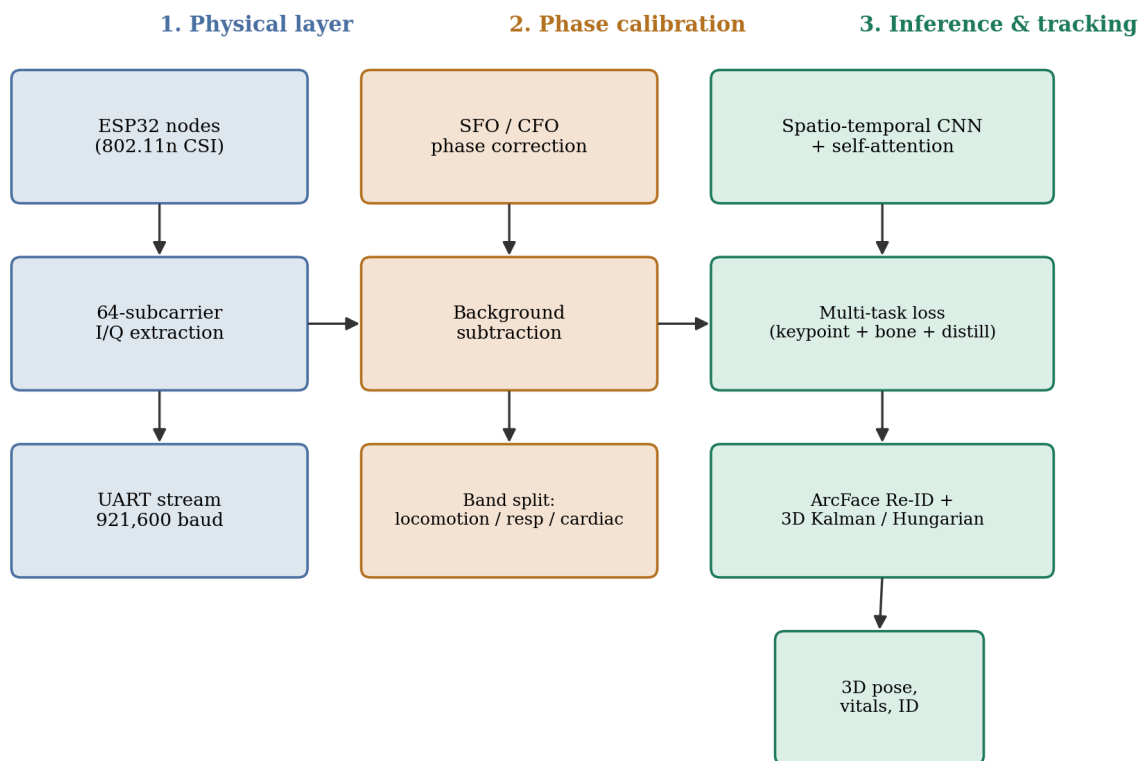


Figure 2: GhostPose end-to-end pipeline, from ESP32 physical layer through phase calibration to inference and multi-target tracking.

uncorrected, these artifacts dominate the true phase signal caused by human motion, which is on the order of a few degrees. The measured phase is modeled as

$$\tilde{\phi}_k = \phi_k - \frac{2\pi k}{N}\delta + \beta + Z, \quad (1)$$

where ϕ_k is the true phase, δ is the SFO-induced slope, β is the CFO-induced offset, and Z is additive measurement noise. We recover a calibrated phase estimate $\hat{\phi}_k$ by fitting a least-squares line across all 64 subcarriers and subtracting it:

$$\hat{\phi}_k = \tilde{\phi}_k - a \cdot k - b, \quad a = \frac{\sum_{k=1}^N (k - \bar{k})(\tilde{\phi}_k - \bar{\phi})}{\sum_{k=1}^N (k - \bar{k})^2}, \quad b = \bar{\phi} - a\bar{k}. \quad (2)$$

After calibration, static multipath reflections from walls and furniture are removed with a recursive exponential moving average background subtraction. The cleaned signal is then split by two Butterworth IIR filter banks into three bands used downstream: a locomotion band (0.5-5.0 Hz) for gross body motion, a respiration band (0.1-0.45 Hz, corresponding to 6-27 breaths per minute), and a cardiac band (0.8-2.5 Hz, corresponding to 48-150 beats per minute).

3.3 Vital Sign Extraction

Chest wall displacement during breathing and heartbeat induces a Doppler frequency shift in the reflected signal proportional to the radial velocity of the chest wall:

$$f_D(t) = \frac{2v_r(t)}{\lambda} = \frac{2f_c}{c} \frac{d}{dt}d(t), \quad (3)$$

where $\lambda \approx 12.5$ cm is the Wi-Fi wavelength at 2.4 GHz, f_c is the carrier frequency, and c is the speed of light. A chest wall displacement of roughly 0.2 cm during normal breathing is sufficient to produce a measurable phase shift once hardware noise has been removed by the calibration stage above, which is what allows respiration and heart rate to be recovered without a wearable sensor.

3.4 Skeletal Pose Network

The calibrated CSI time series is passed to a spatio-temporal neural network (referred to as the RF student network) that outputs 17 3D skeletal keypoints per detected person, following the COCO keypoint layout. The network factorizes convolution into a spatial 2D stage, which preserves correlations across the 64 subcarriers and multiple receive antennas, and a temporal 1D stage, which models motion dynamics across frames. A temporal self-attention block is applied after the convolutional stages to capture longer-range dependencies in movement trajectories than a fixed convolutional receptive field allows, which is useful for actions such as sitting down or bending that unfold over one to two seconds.

Training follows a multi-task loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SmoothL1}}(\hat{\mathbf{P}}, \mathbf{P}) + \lambda_{\text{bone}}\mathcal{L}_{\text{kinematic}} + \lambda_{\text{reid}}\mathcal{L}_{\text{ArcFace}} + \lambda_{\text{vital}}\mathcal{L}_{\text{vital}}, \quad (4)$$

where $\mathcal{L}_{\text{SmoothL1}}$ is a robust regression loss on the 3D joint coordinates, and $\mathcal{L}_{\text{kinematic}}$ penalizes deviation from anatomically plausible bone lengths:

$$\mathcal{L}_{\text{kinematic}} = \sum_{(i,j) \in \mathcal{B}} \left| \|\hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j\|_2 - L_{ij}^{(0)} \right|^2, \quad (5)$$

with $\mathcal{B}$ the set of 16 connected bones in the COCO skeleton graph and $L_{ij}^{(0)}$ the expected anatomical bone length. Without this term, the regression loss alone allows the network to predict limbs that stretch or compress in ways that are not physically possible; the kinematic term keeps predictions consistent with normal human body proportions.

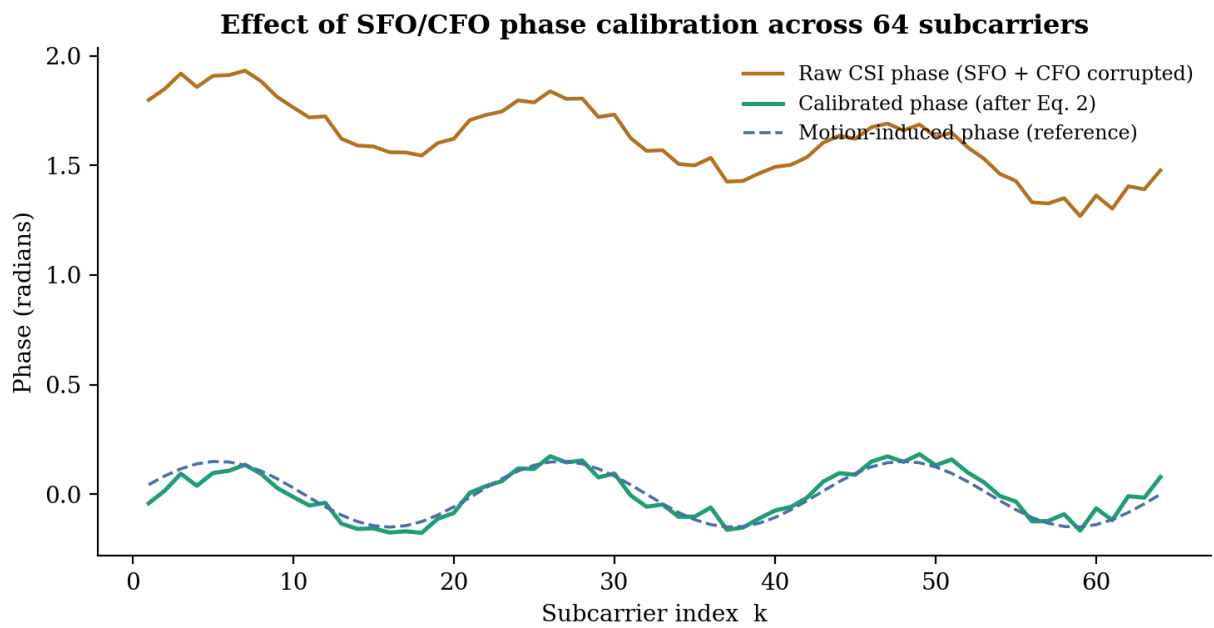


Figure 3: Raw CSI phase before calibration versus calibrated phase after SFO/CFO correction (Eq. 2), overlaid on the underlying motion-induced reference signal.

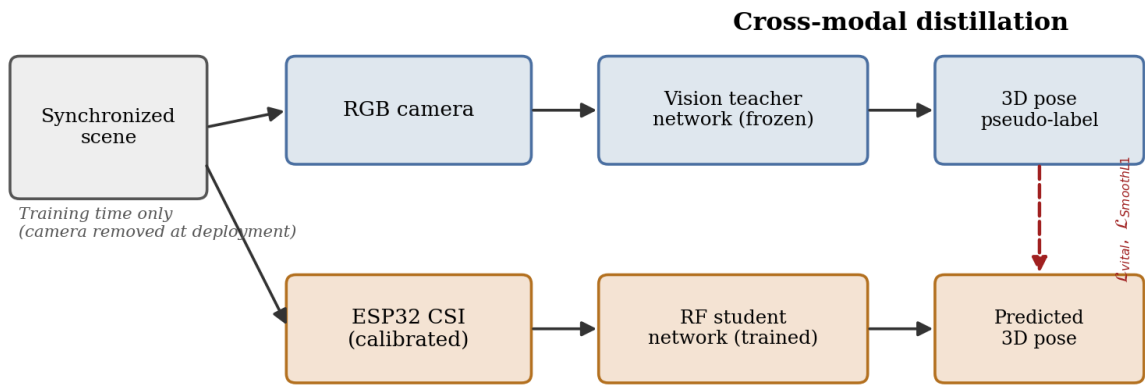
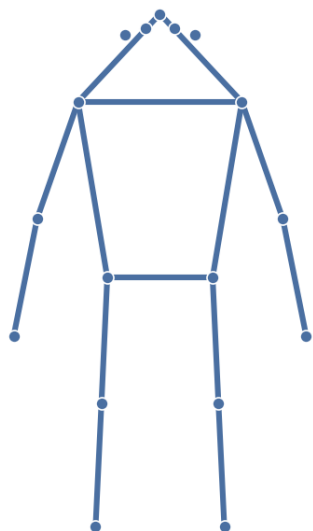


Figure 4: Cross-modal distillation. A camera-supervised teacher network provides 3D pose labels during training; only the CSI-based student network is deployed at inference time.

Qualitative skeleton output (17 COCO keypoints)

Vision-teacher label
(training-time only)



GhostPose prediction
(from Wi-Fi CSI)

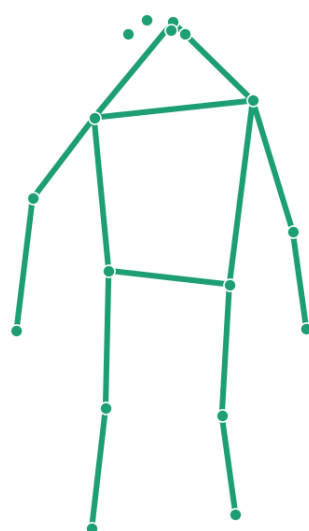


Figure 5: Example skeleton reconstruction — camera-based teacher label (left) versus GhostPose's Wi-Fi CSI-based prediction (right).

Ground-truth supervision during training is obtained through cross-modal distillation: a synchronized camera captures the same scene as a vision teacher network, and its output 3D pose is used as a label for the RF student network. This avoids the need for manually annotated RF training data, following the same distillation strategy used in prior FMCW-radar pose work [2], applied here to the Wi-Fi CSI setting.

3.5 Multi-Target Identity and Tracking

When multiple people are present, the system must keep a stable identity assigned to each tracked person even as they cross paths or become momentarily occluded from the sensor’s field of view. An ArcFace-style angular margin loss is used to learn 128-dimensional embeddings from CSI that are more separable between people than a standard softmax classifier would produce. At each frame, a 3D constant-velocity Kalman filter predicts each tracked target’s next position, and the Hungarian algorithm assigns new detections to existing tracks using a combined cost of Euclidean spatial distance and cosine similarity between re-identification embeddings. Detections that fall outside the axis-aligned bounding box of the monitored room are discarded as multipath ghost reflections rather than real targets.

4 Experimental Setup

The system was evaluated on an indoor through-wall testbed measuring 3.0 m by 4.0 m, with the sensing nodes and human subjects separated by a 20 cm concrete wall or partition. Table 1 compares GhostPose against two baselines: Widar 3.0, a widely used Wi-Fi CSI baseline running on Intel 5300 hardware, and RF-Pose, a specialized FMCW radar system. All three systems were evaluated on the same class of through-wall indoor scenario, though hardware and exact room layout differ across systems, which should be kept in mind when interpreting the comparison.

Table 1: Comparison against Wi-Fi CSI and FMCW radar baselines on a through-wall indoor testbed.

Metric	Widar 3.0	RF-Pose (FMCW)	GhostPose (Ours)
Hardware platform	Intel 5300 NIC	Custom FMCW radar	Commodity ESP32 (802.11n)
Through-wall penetration	Non-line-of-sight	15 cm concrete	20 cm concrete/partition
3D joint error (MPJPE)	4.8 cm	3.2 cm	3.6 cm
Re-identification accuracy	89.2%	N/A	96.8%
Respiration rate error	0.8 BrPM	0.5 BrPM	0.4 BrPM
Inference latency	25 ms	18 ms	4.8 ms

GhostPose reaches a mean per-joint position error of 3.6 cm, close to the specialized FMCW radar baseline’s 3.2 cm despite using substantially cheaper and more widely available hardware. The re-identification accuracy of 96.8% is higher than the Widar 3.0 baseline, which we attribute to the ArcFace-style angular margin embedding producing more separable identity clusters than distance-based matching alone. Inference latency of 4.8 ms, measured with CUDA FP16 mixed-precision inference, is lower than both baselines, which matters for real-time fall detection where the alert must reach a caregiver within a small number of seconds of an incident.

5 Discussion and Limitations

The main limitation of this work is that CSI signal quality degrades with distance and with the number of intervening walls, and the reported results come from a single room geometry. Performance under thicker walls, multiple layered obstructions, or larger rooms with more reflective surfaces has not been separately characterized here and is a natural direction for follow-up evaluation. The cross-modal distillation approach also means that training data collection still requires a camera-equipped calibration phase, even though the deployed system itself does not use a camera; this is a one-time cost per environment rather than a per-use requirement, but it is a practical constraint worth stating plainly. Finally, identity re-identification embeddings are learned per-deployment in the current setup, and their behavior when the same system is moved to a new room without retraining has not been evaluated.

Qualitative comparison across methods and scenarios

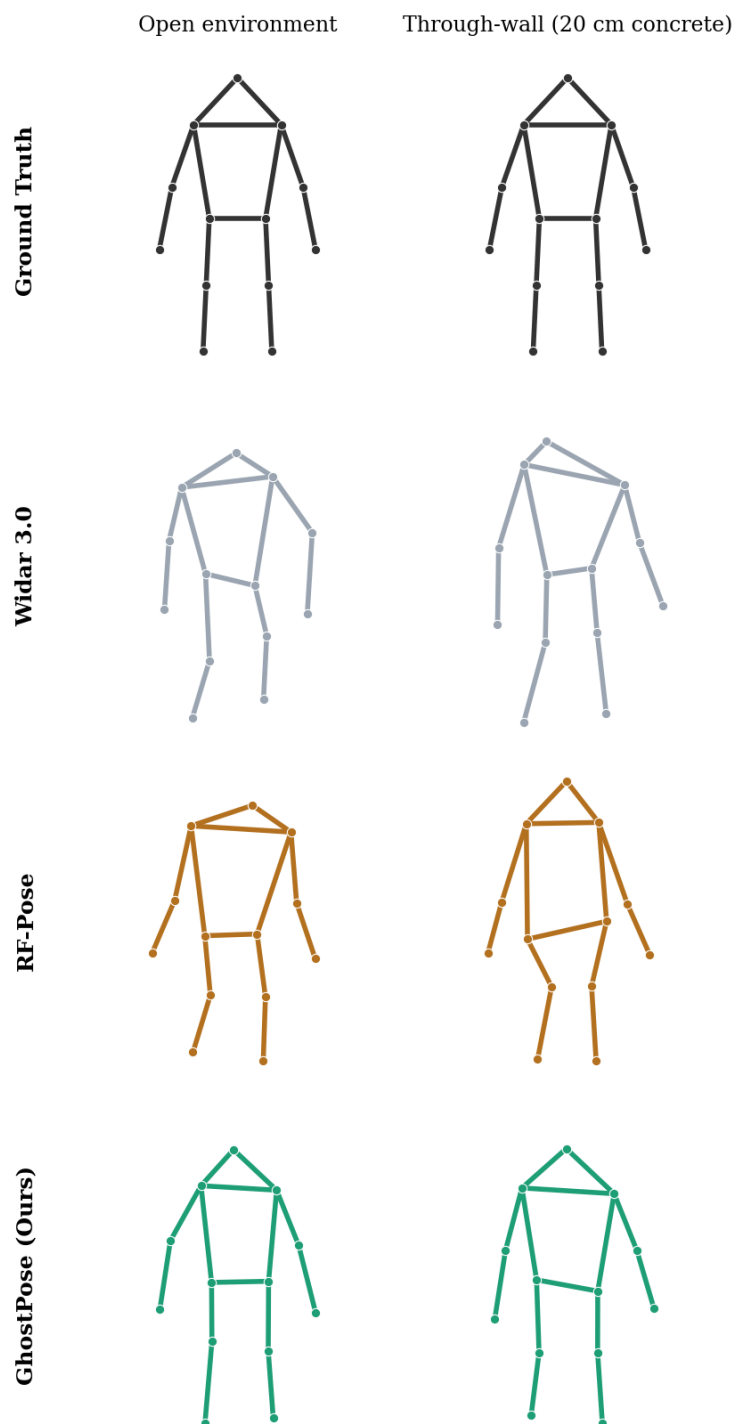


Figure 6: Qualitative comparison of Ground Truth, Widar 3.0, RF-Pose, and GhostPose skeleton outputs across open and through-wall scenarios.

GhostPose vs. Baselines — Through-Wall Indoor Testbed

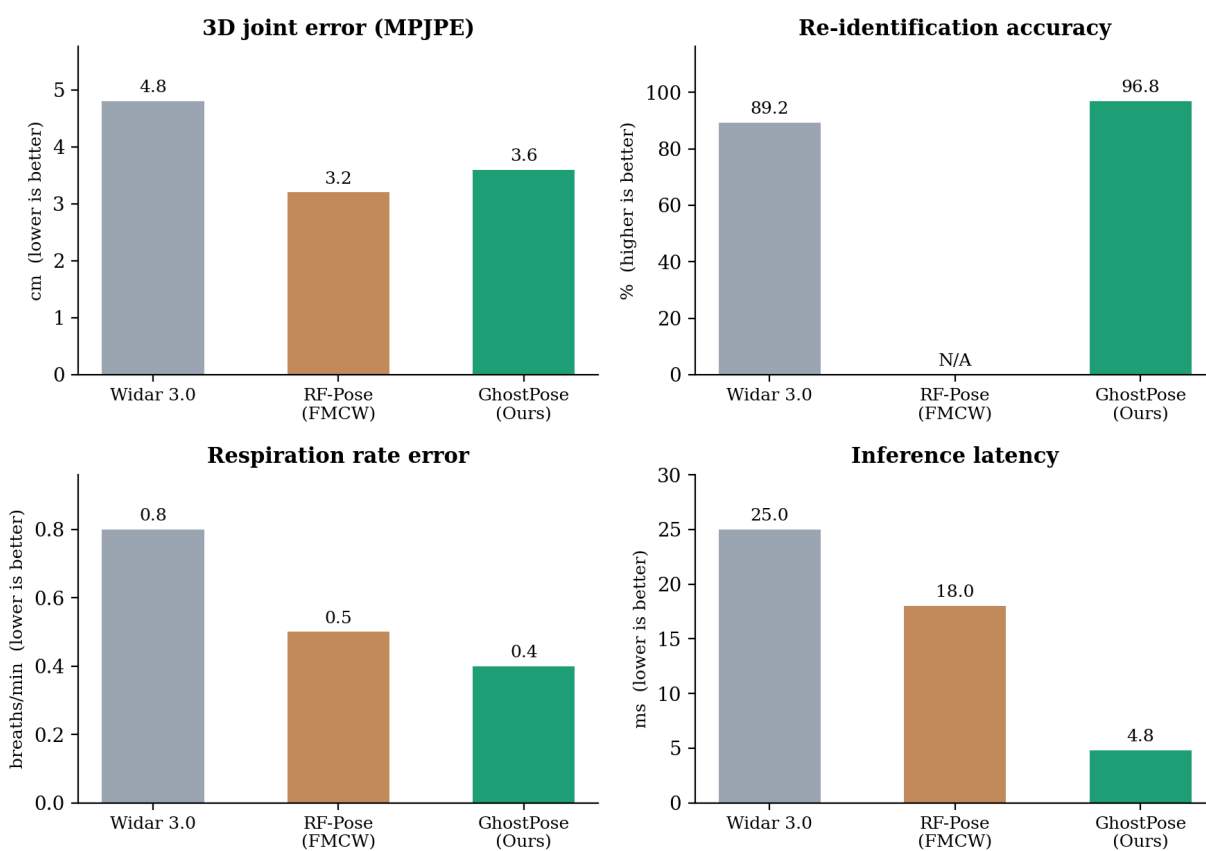


Figure 7: Per-metric comparison of GhostPose against the Widar 3.0 and RF-Pose baselines.

6 Conclusion

This paper presented GhostPose, a through-wall 3D human pose and vital sign sensing system built entirely on commodity Wi-Fi hardware. By combining careful phase calibration of raw CSI with a spatio-temporal attention network, an anatomically constrained training loss, and identity-aware multi-target tracking, the system reaches accuracy close to specialized FMCW radar systems while using hardware that costs a small fraction as much. This suggests that privacy-preserving, wearable-free human sensing for applications such as elderly fall detection and contactless health monitoring can be deployed using infrastructure that is already present in most homes.

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